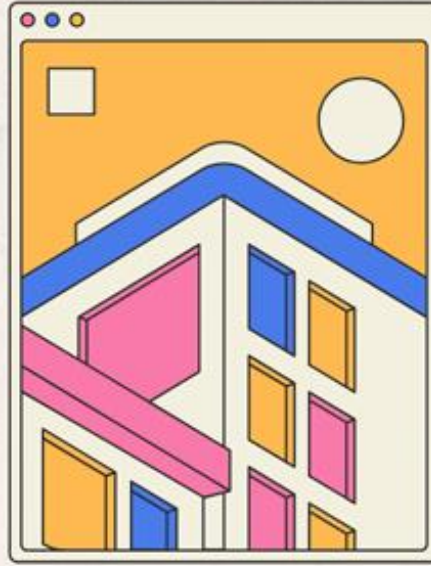
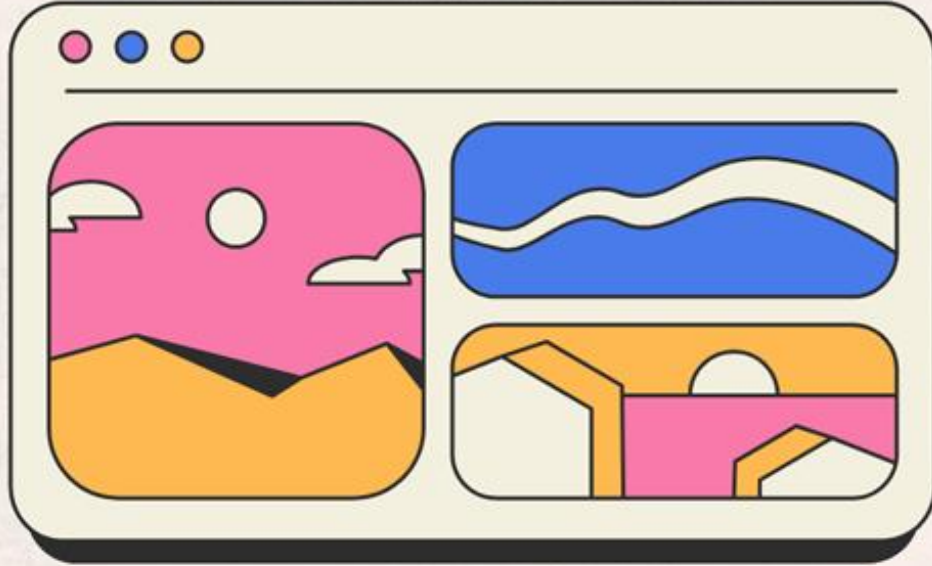
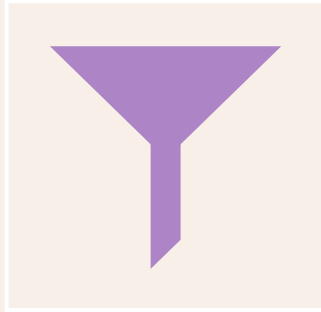


Social Media Marketing



Problem & Proposal



Problem:

No single dashboard to analyze all marketing campaigns.

Data is spread across multiple channels (Email, YouTube, Instagram, Google Ads, Facebook).

Hard to compare campaign types and audience behavior.



Proposal:

Build an interactive Power BI dashboard for multi-channel marketing analytics.

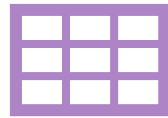
Combine all campaign data in one model.

Provide KPIs such as CTR, ROI, Conversion Rate & Engagement Score.

Tools & Technologies Used



Power BI: Data cleaning, Data modeling (Star Schema), DAX measures, interactive dashboards



Excel: KPI validation, Pivot Tables analysis



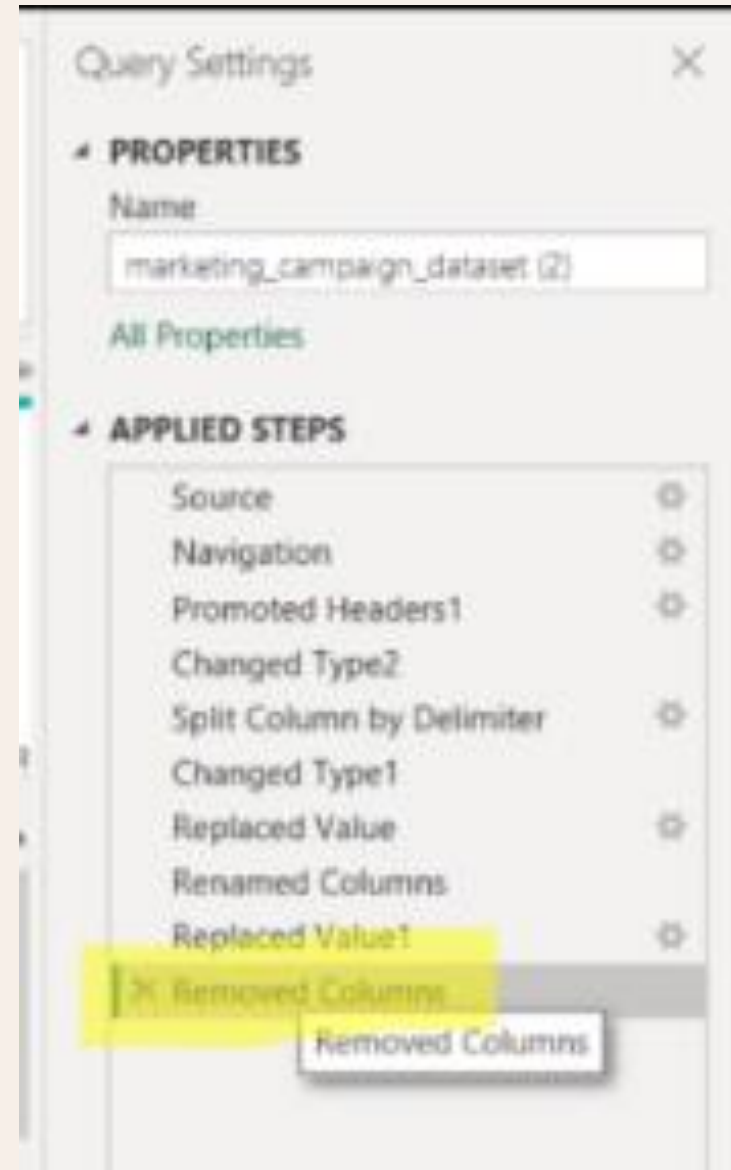
Tableau: Executive overview & campaign visualization



GitHub: Version control & project delivery

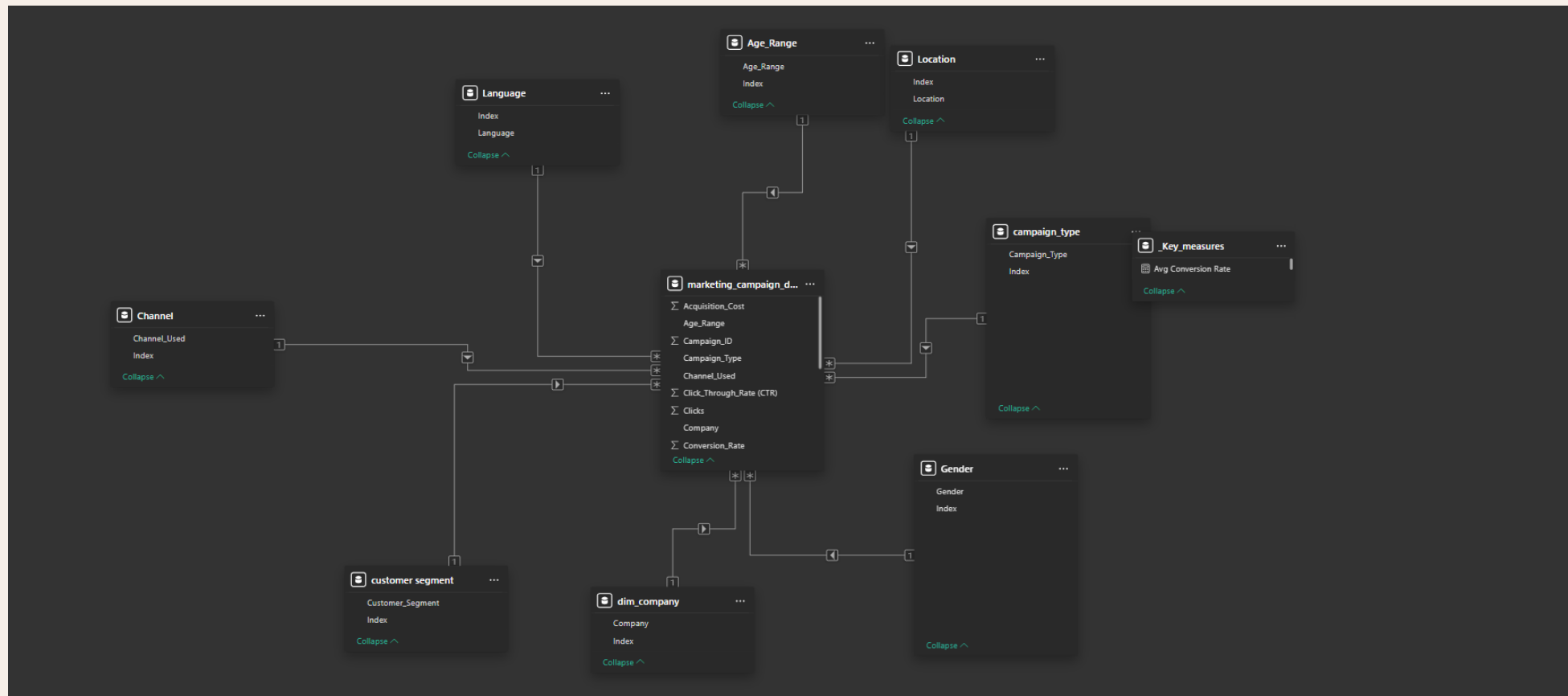
Data Source & Cleaning

- Dataset: Marketing Campaign
- Source: Kaggle
- Cleaning Steps:
 - Handled nulls.
 - Fixed data types (Dates, Numbers).
 - Removed unwanted columns.



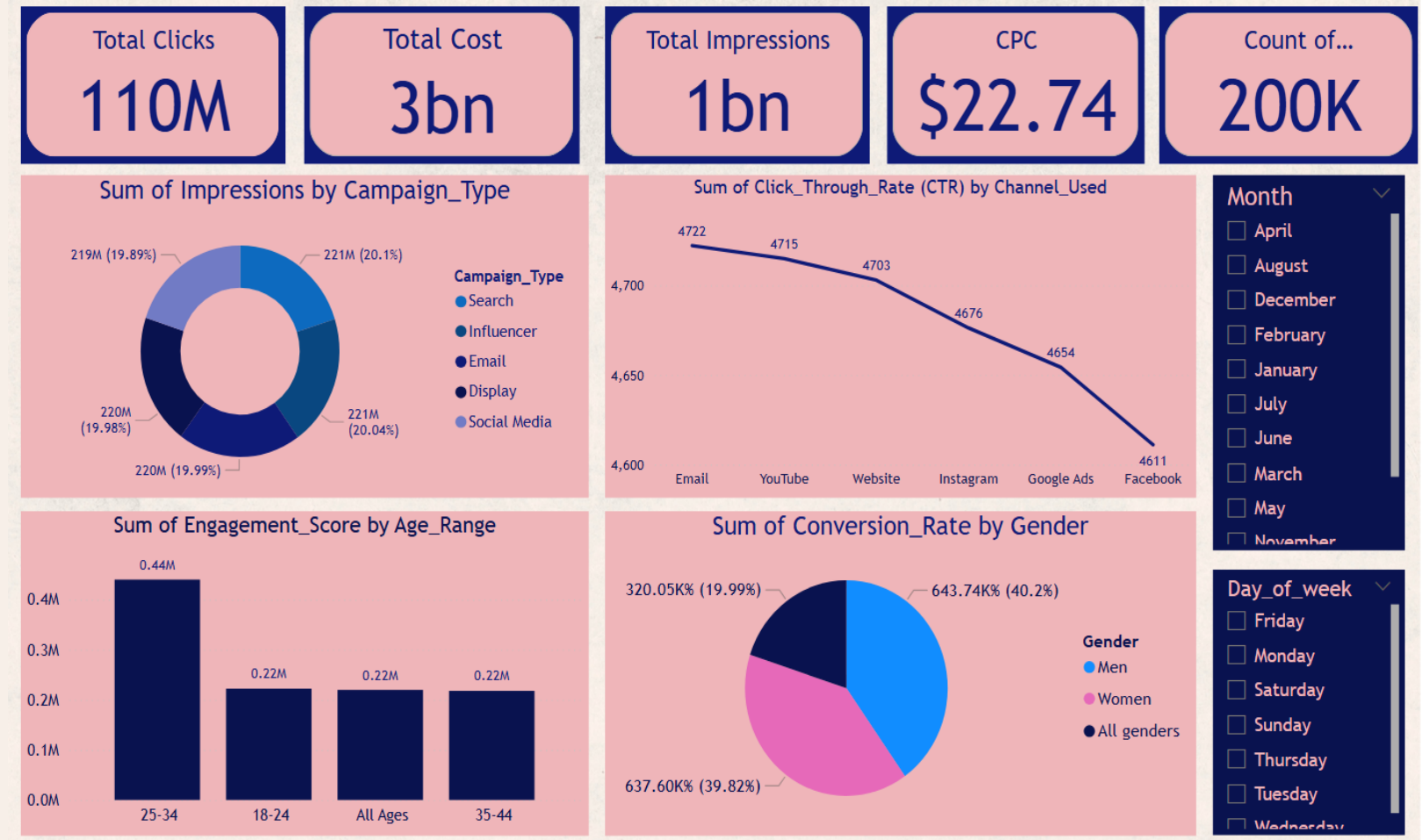
Data Model (Star Schema)

- The central Fact Table is **marketing_campaign_data**, containing all the numeric metrics like cost, clicks, CTR, conversions, etc. Around it we have dimension tables such as Channel, Age Range, Location, Campaign Type, Gender, and Company.



Executive Summary – Power BI

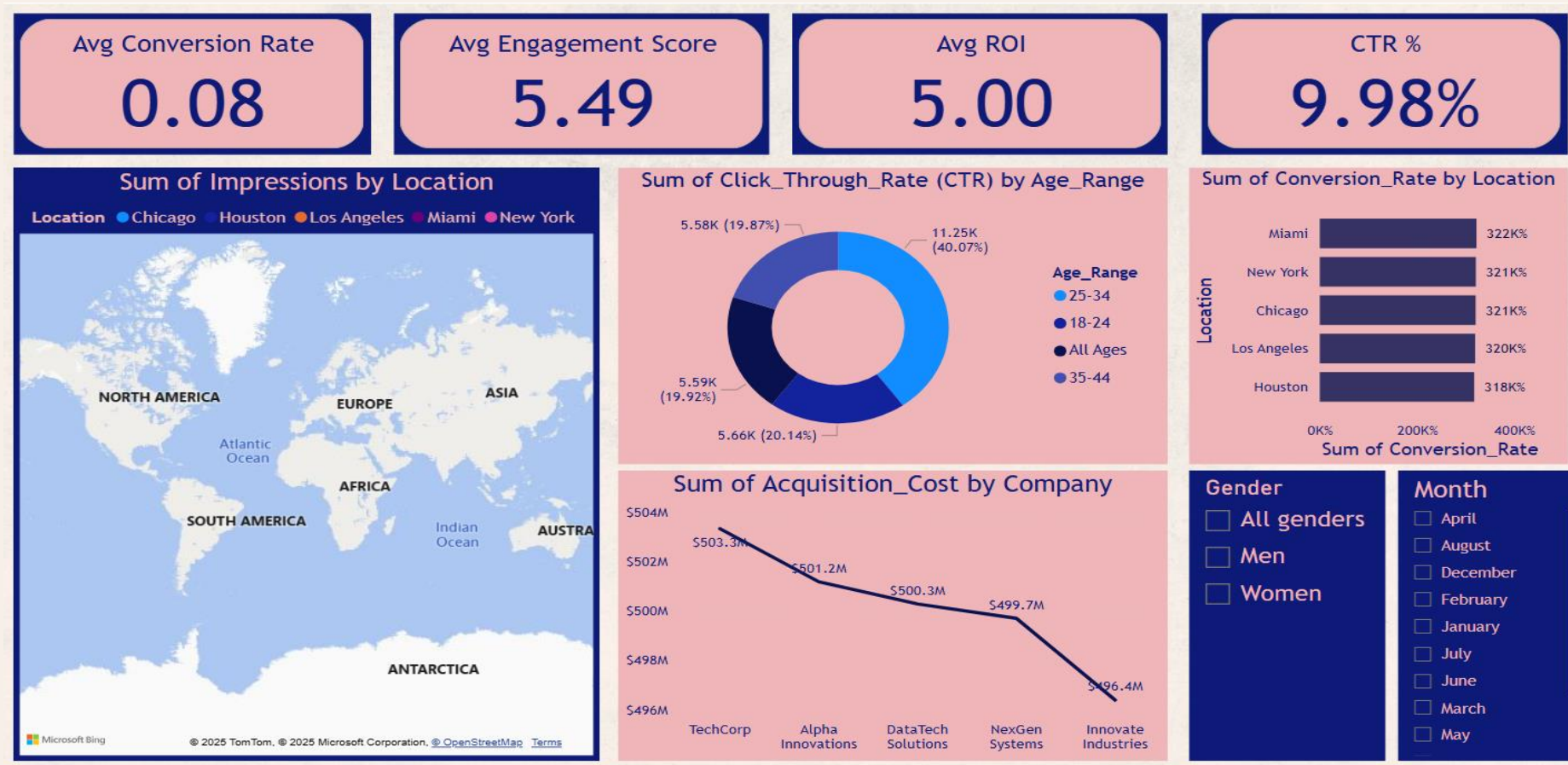
- Shows overall performance: Total Clicks, Total Cost, Total Impressions.
- Includes main KPIs like CTR%, CPC, Avg ROI.
- Provides high-level view for executives.



Campaign Analysis

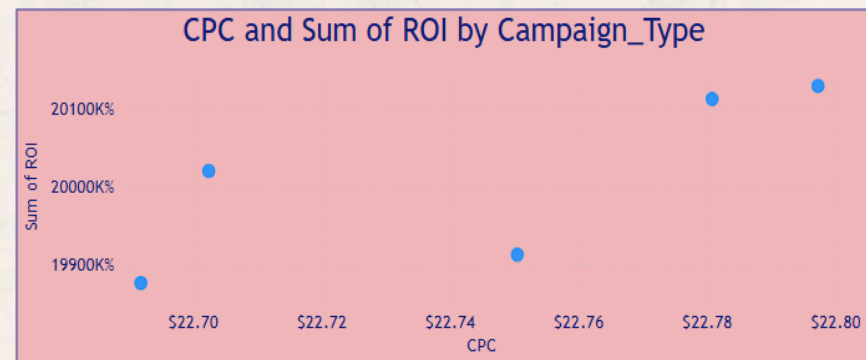
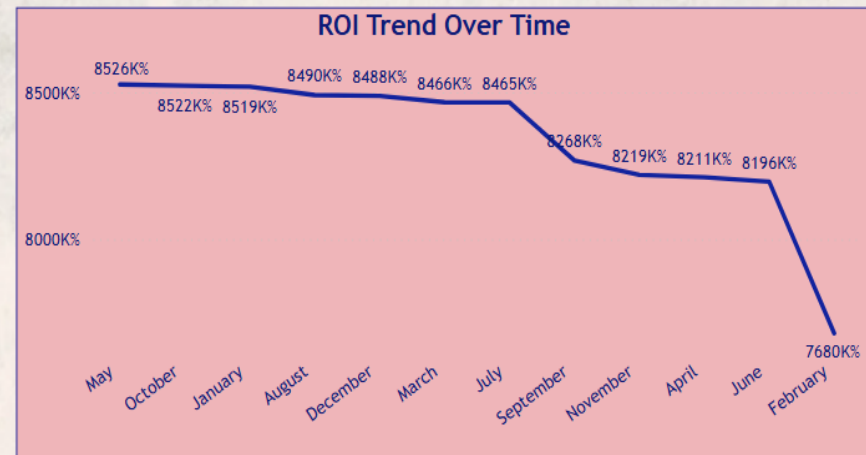
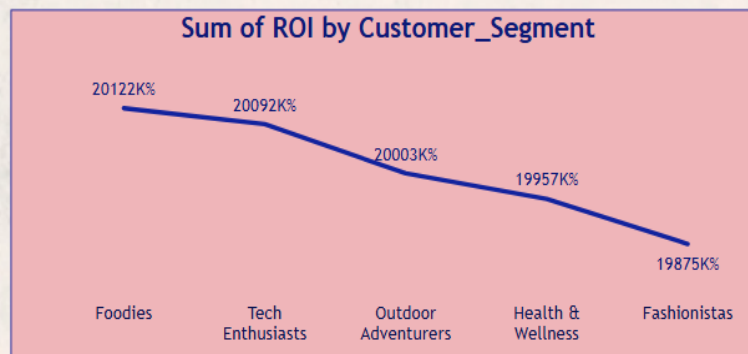
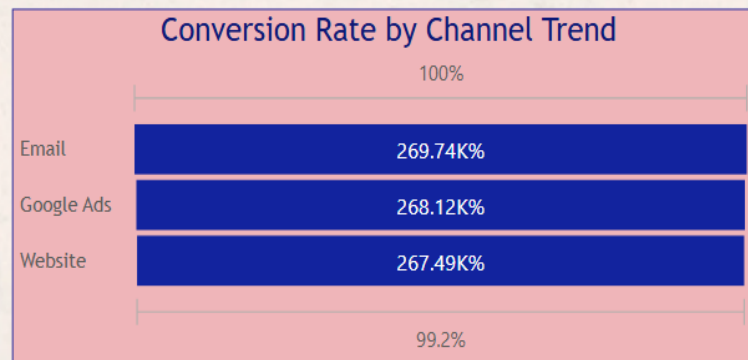
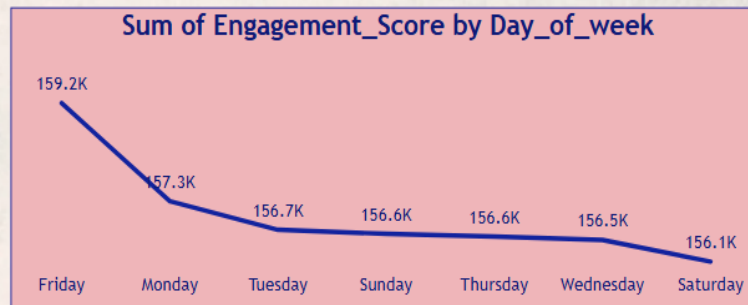
– Power BI

- This analysis shows which campaign types perform best in terms of impressions and which channels drive higher CTR. It helps evaluate the effectiveness of each marketing channel and identify high-performing areas.



Audience – Power BI

- This page shows detailed campaign performance insights: Engagement is highest on Fridays, Email has the strongest conversion rate, ROI decreases over time, and Foodies/Tech Enthusiasts are the top-performing customer segments. The CPC vs ROI chart helps identify which campaign types deliver the best return for the lowest cost.



Age_Range

All

Gender

All

Performance Comparison by Company, Audience & Channel

- All companies show stable ROI around 5.0. The most engaged users are aged 18–34, especially men 18–24 and women 25–34. Display and Influencer channels achieve the highest conversion rates, while all channels maintain consistent ROI around 5.0.

Row Labels	Average of ROI	Average of Conversion_Rate
Alpha Innovati	5.006	8.008%
DataTech Soluti	5.006	7.999%
Innovate Industr	5.002	8.038%
NexGen System	4.991	7.974%
TechCorp	5.007	8.016%
Grand Total	5.002	8.007%

Row Labels	Average of Enga	Sum of Impressions
All Ages	5.487	220,352,342
Men 18-24	5.515	221,226,637
Men 25-34	5.492	220,388,856
Women 25-34	5.493	220,449,734
Women 35-44	5.487	219,042,735
Grand Total	5.495	1,101,460,304

Row Labels	Average of Enga	Sum of Impressions
All Ages	5.487	220352342
Men 18-24	5.515	221226637
Men 25-34	5.492	220388856
Women 25-34	5.493	220449734
Women 35-44	5.487	219042735
Grand Total	5.495	1101460304

Row Labels	Average of Com	Average of ROI
Display	8.009%	5.007
Email	7.979%	4.994
Influencer	8.031%	5.011
Search	8.002%	5.008
Social Media	8.013%	4.992
Grand Total	8.007%	5.002

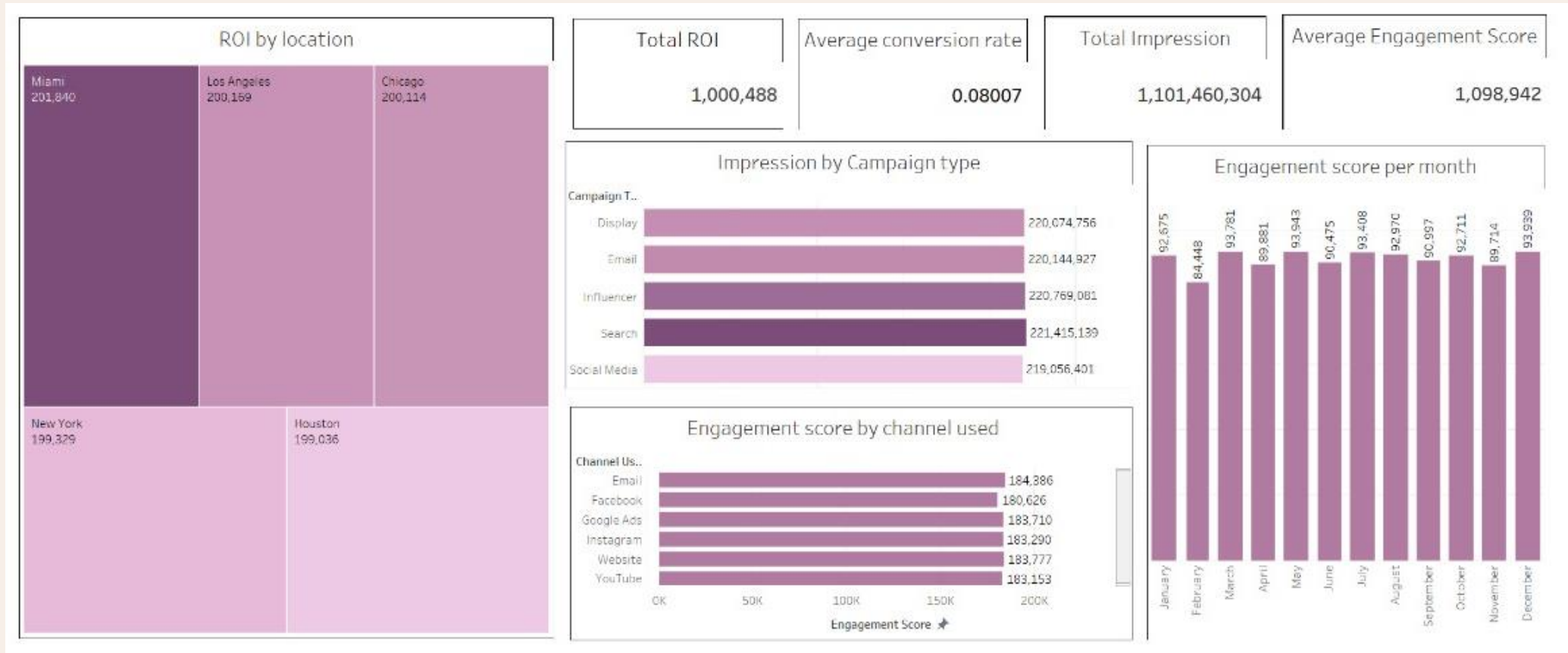
Overall Performance Analysis – Excel

- Campaigns generated strong results with \$2.5B total cost, 110M clicks, 5.0 ROI, and 8% conversion rate. Influencer and Display campaigns perform best, the 18–34 age group is the most engaged, TechCorp leads company ROI, and Facebook shows the weakest channel performance.



Executive Overview – Tableau

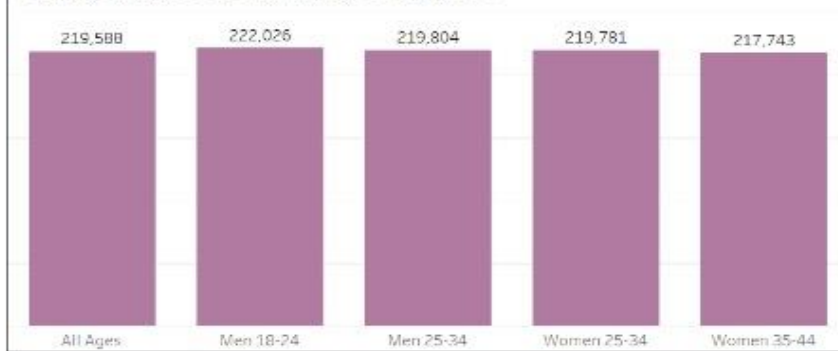
- Miami achieves the highest ROI, Search leads in total impressions, Email is the most engaging channel, and overall engagement remains stable throughout the year. With 1.1B impressions and an 8% conversion rate, the campaigns demonstrate strong profitability and reach.



Campaign Performance – Tableau

- The highest engagement comes from users aged 18–24, Email delivers the strongest ROI, Friday is the peak engagement day, impressions are highest from August to October, and Foodies and Tech Enthusiasts achieve the best conversion rates.

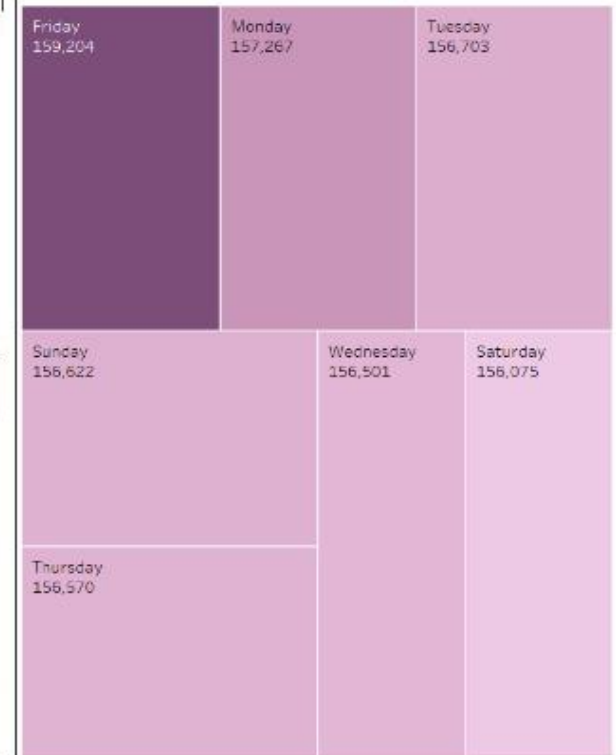
Engagement score by Target audience



ROI by channel used



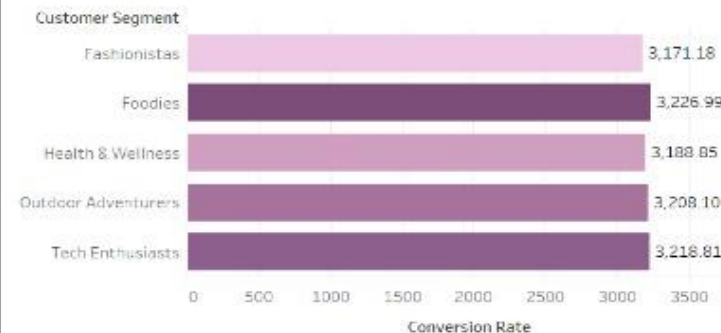
Engagement by day



Impression by month



Conversion rate by Customer segment



Results & Insights

Email is the strongest channel with the highest CTR and conversion rate. Age group 25–34 shows the highest engagement, while Miami and New York lead in impressions and conversions. ROI has slightly declined over time, but Foodies and Tech Enthusiasts remain the most profitable segments. CPC is stable across campaigns, with some campaigns achieving high ROI at relatively low cost.

Challenges & Solutions

Challenge	Solution
Inconsistent data formats	Used Power Query to standardize data
Choosing the right visuals	Used 3-page layout (Executive, Campaign, Audience)
Understanding KPIs formulas	Created DAX measures and validated them

Business Recommendations

- **Increase budget allocation to Email and Influencer campaigns**, as they show the highest ROI and conversion rates.
- **Focus marketing efforts on the 18–34 age group**, as this segment demonstrates the highest engagement and responsiveness.
- **Prioritize high-performing locations** like Miami and New York for future campaign targeting.
- **Optimize or reconsider Facebook campaigns**, as they show relatively lower engagement and CTR.
- **Invest more in high-performing customer segments** such as Foodies and Tech Enthusiasts.
- **Monitor ROI trends over time**, as recent months show a slight decline in efficiency.

Future Work

01

Connect the model to real-time data via APIs.

02

Add predictive analysis for best-performing channels.

03

Automate refresh schedule in Power BI Service.

04

Apply Row-Level Security for different managers.

GitHub & Delivery

[haneen35/Data-analysis-depi-project](#)